

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04 01

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

Total Marks: 100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Soundariga. (192529342) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

 M.

FUNDAMENTAL OF DATA SCIENCE.

A college wants to identify students who may fail in the final exam. How can Data Science help?

Data Science can analyze students' marks, attendance, assignment scores, and previous exam performance. Using this data, a machine learning model can predict which students are likely to fail. Teachers can identify these students early and provide extra classes, guidance and support. Thus, Data Science helps improve student performance and reduce failure rates.

3. An Online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

A. Recommendation System is used for this purpose. It analyzes customers' previous purchase, searches, clicks, and interests. Based on these patterns, it recommends products that the customer may be interested in buying. For ex, shopping websites can suggest products similar to those previously viewed by the customer.

- 5) A Company wants to reduce customer loss. how can data science support this?

Data science can be used for customer churn prediction

It analyzes customer complaints, purchase history, usage patterns, and interaction with the company.

The system can identify customers who are likely to leave the company.

The company can then offer discounts, better services or support to retain those customers.

- 7) A supermarket want to know which products sell more during festivals. how can data science help?

Data science can analyze previous festival sales, customer preference and purchasing patterns.

It can identify which products had high demand during previous festivals. Using this information, the supermarket can predict future demand and maintain sufficient stock.

This helps avoid shortage or excess stock and increases sales.

20
A farmer wants to improve crop yield. Give one use of Data Science.

Data Science can analyze soil conditions, rainfall, temperature, weather and crop history. Using this information, it can help farmers select suitable crops for a particular area. It can also predict the best time for irrigation, fertilizer application and harvesting. This can increase crop productivity and reduce unnecessary resource usage.

11. A weather app updates temperature every minute. Which facet of data is involved?

The Big Data characteristic involved is velocity. Velocity refers to the speed at which data is generated, collected and processed. In a weather application, temperature and other weather information are continuously updated.

* Therefore, frequent and real-time data updates represent high data velocity.

18 Customer Comments on social media are collected for analysis. What type of data is this?

Social media Comments are generally considered Unstructured Data. They do not follow a fixed table format like traditional databases.

* They may contain text, emojis, hashtags, images, videos and other information. Data Science techniques such as Natural language processing can be used to analyze such data.

19 A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present.

The dataset contains incomplete and inaccurate data. Missing age values represent the problem of incompleteness, because some required information is absent. Incorrect phone numbers represent inaccuracy, because the stored information is not correct.

A company stores large files in Hadoop distributed file system. Which part of the Big data ecosystem is used?

The Component used is HDFS - Hadoop Distributed file system.

* HDFS is storage system in Hadoop ecosystem designed to store very large datasets.

* It divides large files into blocks and stores them across multiple computer in a cluster. This provides distributed storage and improved reliability and scalability.

17. An Organization wants to collect live sensor data from IoT device, which Big Data ecosystem Component is suitable?

Apache Kafka is suitable for collecting and handling real-time streaming data. IoT device can continuously generate sensor data such as temperature, pressure and location.

Kafka receives and broadcasts this data efficiently to other systems for processing and analysis. real-time data streaming.